

QR code Hostel Dining Management System

A Micro SaaS solution for Smart Hostel Meal Access

Domain: Education

Team name: Byte Tech

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Introduction

Manual verification of hostel students.

Unauthorized students entering the mess.

No digital attendance records.

Long queues during meal times.

Duplicate meal usage.

Difficult to generate meal reports.



Problem Statement

Students have QR-enabled college IDs but hostel operations aren't integrated

Result: slow verification, unauthorized access, duplicate meals, poor tracking

Goal: replace manual checks with an automated, auditable flow

Proposed Solution

Cloud Micro-SaaS

Centralized
student registry

QR Scanning

Warden mobile
scanner validates
IDs

Reporting

Block duplicates,
record
attendance

Real-time Enforcement

Attendance logs, meal
analytics, exportable
reports

Objectives



Reduce Wait Time

Instant verification at entry



Prevent Unauthorized Access

Verify hostel registration and eligibility



Maintain Records

Persistent meal attendance per student



Generate Analytics

Dashboards and scheduled reports for admins

Existing VS Proposed – At a Glance

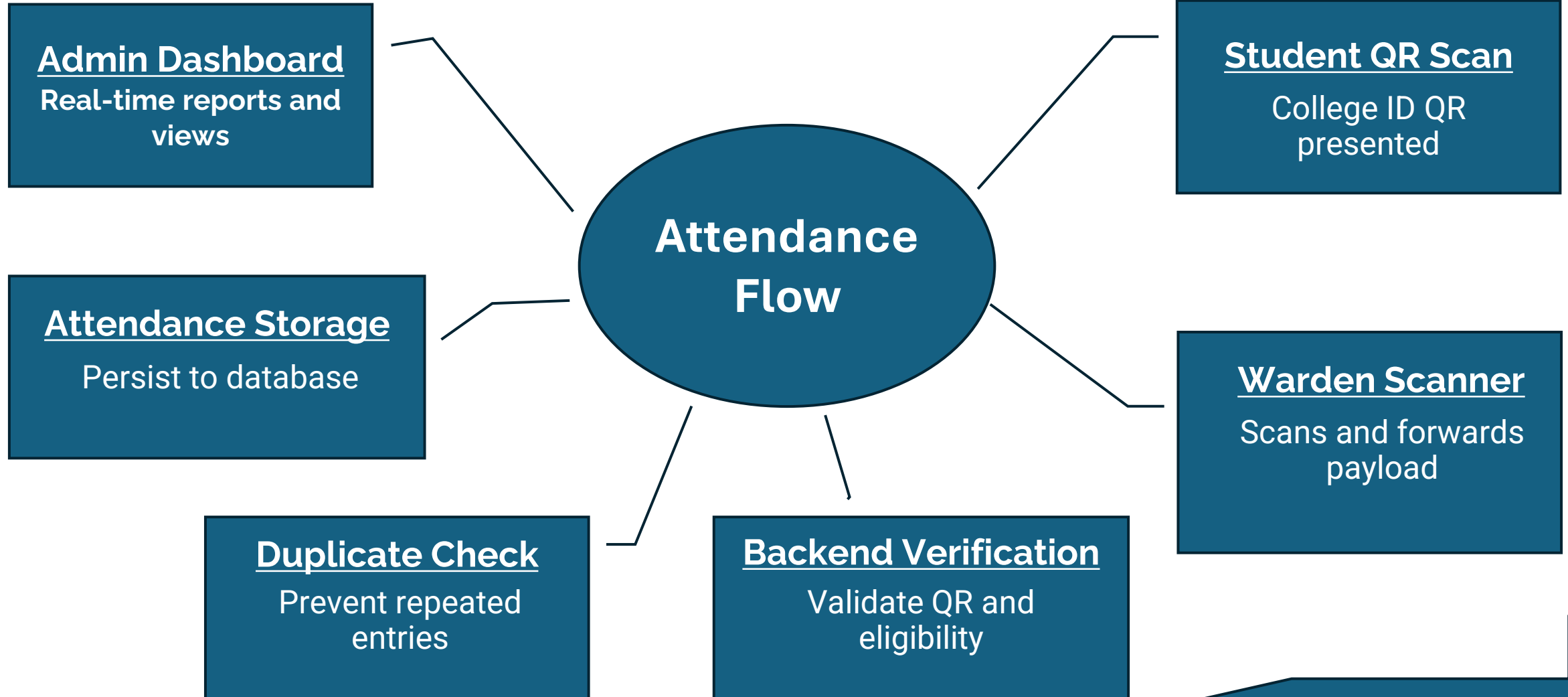
Existing System

- Manual verification
- Paper records
- No reports
- Slow, error-prone

Proposed System

- QR scan verification
- Digital database
- Automated reports
- Fast, auditable

System Architecture – High Level



Operational Workflow

- | | |
|------------|--|
| 1 - Entry | Student arrives at mess |
| 2 - Scan | Warden scans college ID QR with mobile scanner |
| 3 - Verify | Check hostel registration , meal , plan , and duplicates |
| 4 - Record | Attendance logged ; access allowed or denied |
| 5 - Report | Dashboard reflects live counts ; admin exports reports |

Features by role

Student

- Use existing QR ID
- No extra app
- Faster entry

Warden

- Mobile QR scanner (Flutter)
- Instant access decision
- View recent attendance

Admin

- Register students & hostels
- Generate analytics & exports
- Manage users & permissions

Tech & Data Model



Tech Stack

- Frontend: React.js
- Mobile Scanner: Flutter + ZXing
- Backend: Spring Boot, JWT security
- DB: MySQL; Cloud: AWS/Azure/Render



Database Design

- Tables: Student, Hostel, QR_Mapping, Meal_Attendance, Users, Reports
- Relationships: Student → QR_Mapping → Meal_Attendance
- Indexes for fast duplicate detection and daily reports



Security & Ops

- JWT auth for APIs
- Role-based access (warden/admin)
- Audit logs and exportable CSVs for compliance

Future Enhancements



Face Recognition

Optional biometric layer for contactless verification and fraud reduction



Parent Notifications

Real-time alerts for meal usage, low balance, and attendance summaries.



Mess Menu App

Daily menus, allergens, and student feedback – improves satisfaction and planning



Online Hostel Registration

Self-serve onboarding: room allocation, dietary preferences, and documents



Fee Verification

Automated cross-checks with payment records to block ineligible access.



AI Meal Demand Prediction

Forecast demand by day/time to reduce waste and optimize procurement.



Integration with College ERP

Sync student records, fee status, and attendance for a single source of truth.

Bussiness Model

Target Market

- Colleges & Universities
- Polytechnic Institutions
- Private Hostels
- Educational Campuses

Our Value

- Smart dining using existing student ID cards
- Faster, secure meal verification
- Seamless ERP integration
- Reduced food wastage & operational costs
- Scalable cloud-based solution

Revenue Model

- Subscription Plans:** Free | Basic | Premium | Enterprise
- ERP Integration Services
- Training & Support
- Analytics & AMC

Growth Strategy

- Launch with pilot institutions
- Partner with leading ERP providers
- Expand across India
- Future AI-powered dining analytics

Subscription Model

Plan	Price	Best For	Key Features
Free	₹0	Pilot & Small Hostels	Up to 100 students, Basic meal tracking
Basic	₹4,999/Month	Small & Medium Colleges	Up to 1,000 students, ERP Sync, Reports
Premium	₹9,999/Month	Large Institutions	Unlimited students, Analytics, Notifications
Enterprise	Custom Pricing	Universities & Multi-Campus	Custom ERP Integration, API Access, Dedicated Support



Conclusion

Seamless Access

Students use existing QR IDs — zero new hardware for users.

Fraud Prevention

Controls for unauthorized use (QR + optional biometrics).

Operational Efficiency

Automated records, fewer forms, faster service lines.

Actionable Insights

Real-time analytics and reports to optimize budgets and meal planning.

